9

First-order logic: Syntax

Terms

Formulae

Construction trees

Logic in Computer Science Lab
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Based on prof. Tamás Kádek's lectures

First-Order Language

An ordered 5-tuple $\mathcal{L}^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ defines the language of the first-order (predicate) logic, where

1. $LC = \{\neg, \supset, \wedge, \vee, \equiv, (,), =, \exists, \forall, ,\}$ is the set of *logical constants*,
2. Var is the countable infinite set of variables,
3. $Con \neq \emptyset$; is the set of *non-logical constants* such that $LC \cap Con = \emptyset$, *and*

$$Con = \bigcup_{i=0}^{\infty} (\mathcal{P}(i) \cup \mathcal{F}(i));$$

4. $Term$ is the set of *terms*;
5. $Form$ is the set of *formulae*;

where LC, Var, $\mathcal{P}(i)$ and $\mathcal{F}(i)$ are pairwise disjoint.

The LC, Var and Con sets contains the letters of the alphabet, while the terms and formulas as sequences of letters are the words of the first-order language.

1. Logical constant set

- the unary connective:
  - $\neg$ negation sign
- the binary connectives:
  - $\wedge$ conjunction sign
  - $\vee$ disjunction sign
  - $\supset$ implication sign
  - $\equiv$ sign of material equivalence
- the quantifiers:
  - $\exists$ existential quantifier
  - $\forall$ universal quantifier
- the equal sign: $=$
- the parentheses: (and)
- the separator sign: ,

First-order language

Non-logical constants set

The symbols of the Con set can be divided into two parts.

$$Con = \bigcup_{i=0}^{\infty} (\mathcal{P}(i) \cup \mathcal{F}(i))\,;$$

1. symbols used as name of functions:
   - $\mathcal{F}(0)$ contains the name parameters,
   - $\mathcal{F}(k)$ where $k \geq 1$ contains the function parameters.
2. symbols used as name of predicates:
   - $\mathcal{P}(0)$ contains the propositional parameters,
   - $\mathcal{P}(k)$ where $k \geq 1$ contains the predicate parameters.

The argument k in $\mathcal{F}(k)$ and $\mathcal{P}(k)$ describes the arity (number of arguments) of the function and predicate parameters.

First-order terms

1. symbols used as name of functions:
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2. symbols used as name of predicates:
   - $\mathcal{P}(0)$ contains the propositional parameters,
   - $\mathcal{P}(k)$ where $k \geq 1$ contains the predicate parameters.

The set of $Term$ is defined inductively:

1. $Var \subseteq Term$;
2. $\mathcal{F}(0) \subseteq Term$;
3. if $f \in \mathcal{F}(k)$ for some $k \geq 1$ and $t_i \in Term$ for all $i = 1, \ldots, k$, then

$$f(t_1, t_2, \ldots, t_k) \in Term.$$

First-order formula

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   - $\mathcal{F}(0)$ contains the name parameters,
   - $\mathcal{F}(k)$ where $k \geq 1$ contains the function parameters.
2. symbols used as name of predicates:
   - $\mathcal{P}(0)$ contains the propositional parameters,
   - $\mathcal{P}(k)$ where $k \geq 1$ contains the predicate parameters.

The set of $Form$ is defined inductively:

Atomic formula

1. $\mathcal{P}(0) \subseteq Form$;
2. $(t_1 = t_2) \in Form$ if
   - t_1 and t_2 are terms ($t_1 \in Term$ and $t_2 \in Term$)
3. $f(t_1, t_2, \ldots, t_k) \in Form$ if
   - $f \in \mathcal{P}(k)$ for some $k \geq 1$ and
   - $t_i \in Term$ for all $i = 1, \ldots, k$.

First-order formula

1. symbols used as name of functions:
   - $\mathcal{F}(0)$ contains the name parameters,
   - $\mathcal{F}(k)$ where $k \geq 1$ contains the function parameters.
2. symbols used as name of predicates:
   - $\mathcal{P}(0)$ contains the propositional parameters,
   - $\mathcal{P}(k)$ where $k \geq 1$ contains the predicate parameters.

1. $Var \subseteq Term$;
2. $\mathcal{F}(0) \subseteq Term$;
3. if $f \in \mathcal{F}(k)$ for some $k \geq 1$ and $t_i \in Term$ for all $i = 1, \ldots, k$, then

$$f(t_1, t_2, \ldots, t_k) \in Term.$$

Non atomic (compound) formula

4. $\neg A \in Form$ if A is a formula ($A \in Form$);
5. $(A \circ B) \in Form$ if
   - A and B are formulas ($A \in Form$, $B \in Form$) a
   - $\circ$ is a binary connective ($\circ \in \{\wedge, \vee, \supset, \equiv\}$);
6. $QxA \in Form$ if
   - Q is a quantifier ($Q \in \{\exists, \forall\}$),
   - x is a variable ($x \in Var$) and
   - A is a formula ($A \in Form$).

Direct subterm

The *direct subterm* of the term is defined as follows:

- if $t \in Var$ then it has no direct subterm,
- if $t \in \mathcal{F}(0)$ then it has no direct subterm,
- if $f(t_1, t_2, \ldots, t_k) \in Term$ then the direct subterms are $t_1, t_2, \ldots, t_k$.

The *set of subterm* of a term $t \in Term$ is a set of terms denoted by $ST(t)$ and is defined inductively as follows:

1. $t \in ST(t)$;
2. if $t' \in ST(t)$ and t'' is a direct subterm of t', then $t'' \in ST(t)$.

Note that ST is a function such that:

$$ST : Term \rightarrow 2^{Term} \setminus \{\emptyset\}$$

1. symbols used as name of functions:
   - $\mathcal{F}(0)$ contains the name parameters,
   - $\mathcal{F}(k)$ where $k \geq 1$ contains the function parameters.
2. symbols used as name of predicates:
   - $\mathcal{P}(0)$ contains the propositional parameters,
   - $\mathcal{P}(k)$ where $k \geq 1$ contains the predicate parameters.

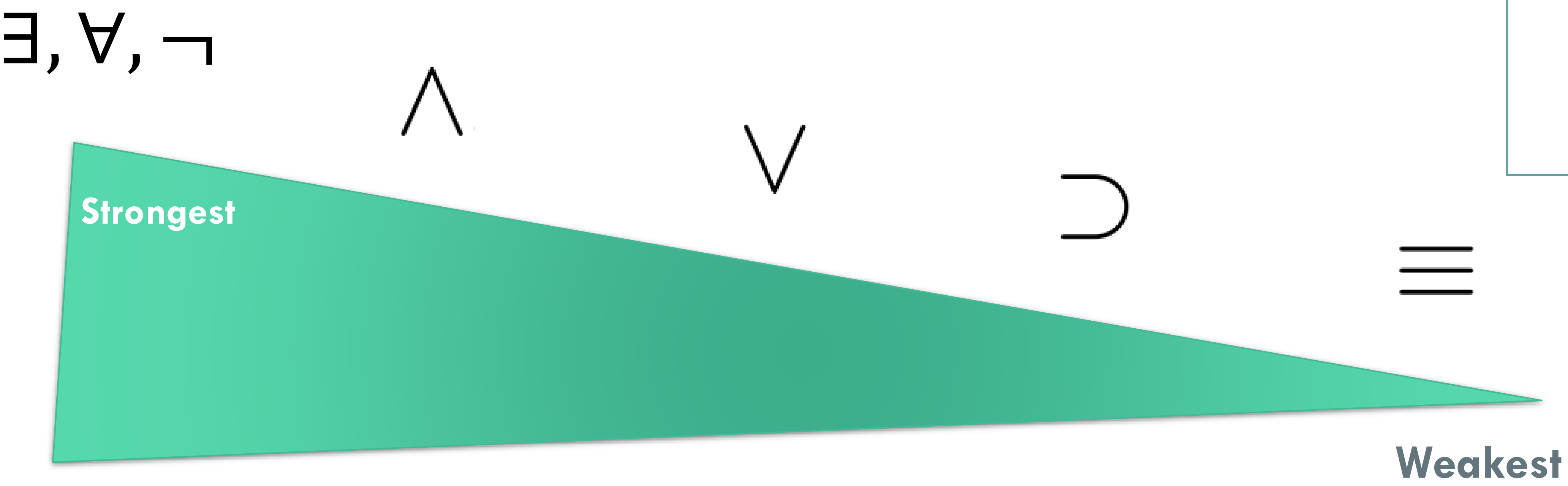
1. $Var \subseteq Term$;
2. $\mathcal{F}(0) \subseteq Term$;
3. if $f \in \mathcal{F}(k)$ for some $k \geq 1$ and $t_i \in Term$ for all $i = 1, \ldots, k$, then

$$f(t_1, t_2, \ldots, t_k) \in Term.$$

Precedence

Logical constants:

- ¬ Negation
- ∀ Forall
- ∃ There_exist
- ∨ Disjunction
- ∧ Conjunction
- ⊃ Implication
- ≡ Equivalency



Parentheses can be dropped when the structure of an expression can be determined by precedence.

Direct subformulae

The *direct subformula* of a formula is defined as follows:

- If $A \in Con$ then the A atomic formula has no direct subformula;
- if $B \in Form$ and $C \in Form$ are arbitrary formulas and $x \in Var$, then
  - the formula $\neg B$, $\exists x B$ and $\forall x B$ have one direct subformula: the formula B;
  - the formulae $(B \wedge C)$, $(B \vee C)$, $(B \supset C)$ and $(B \equiv C)$ have two direct subformulae: the formula B and the formula C.

The *set of subformulae* of a formula $A \in Form$ is set of formulae denoted by $SF(A)$ and is defined inductively as follows:

1. $A \in SF(A)$;
2. if $B \in SF(A)$ and C is a direct subformula of B, then $C \in SF(A)$.

Note that SF is a function such that:

$$SF : Form \to 2^{Form} \setminus \{\emptyset\}$$

Construction tree

The construction tree of a formula $A \in From$ is the finite ordered binary tree defined as follows:

- each nodes are labelled by a formula,
- their root is labelled by the formula A,
- a nodes labelled by $\neg B$, $\exists x B$ and $\forall x B$ has only one child labelled by B,
- a node labelled by $(B \wedge C)$, $(B \vee C)$, $(B \supset C)$ or $(B \equiv C)$ has exactly two children labelled by B and C,
- the leaves labelled by atomic formulae.

The construction tree of a term $t \in Term$ is the finite ordered tree defined as follows:

- each nodes are labelled by a term,
- their root is labelled by the term t,
- a nodes labelled by $f(t_1, t_2, \ldots, t_k)$ has exactly k children labelled by t_1, t_2 and so on until t_k,
- the leaves labelled by atomic terms.

Construction tree

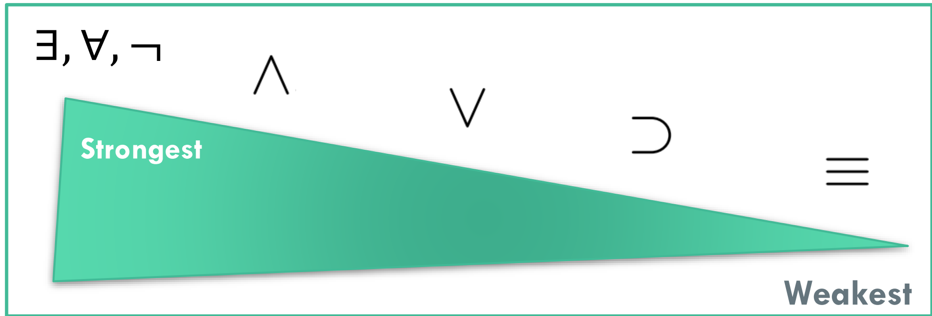
23. Give the construction tree of formula $P(c) \supset P(c) \vee c = d \equiv \neg P(c) \wedge P(d) \wedge P(c) \wedge c = g(c, f(c)) \supset d = c$

- $P(c) \supset P(c) \vee c = d \equiv \neg P(c) \wedge P(d) \wedge P(c) \wedge c = g(c, f(c)) \supset d = c$
  - $P(c) \supset P(c) \vee c = d$
    - $P(c)$
    - $P(c) \vee c = d$
      - $P(c)$
      - $c = d$
  - $\neg P(c) \wedge P(d) \wedge P(c) \wedge c = g(c, f(c)) \supset d = c$
    - $\neg P(c) \wedge P(d) \wedge P(c) \wedge c = g(c, f(c))$
      - $\neg P(c)$
        - $P(c)$
      - $P(d) \wedge P(c) \wedge c = g(c, f(c))$
        - $P(d)$
        - $P(c) \wedge c = g(c, f(c))$
          - $P(c)$
          - $c = g(c, f(c))$
    - $d = c$

Construction tree

25. Give the construction tree of formula $\forall x P(x) \wedge Q(f(c), g(c, g(c, d))) \vee \exists x P(x)$!

- $\forall x P(x) \wedge Q(f(c), g(c, g(c, d))) \vee \exists x P(x)$
  - $\forall x P(x) \wedge Q(f(c), g(c, g(c, d)))$
    - $\forall x P(x)$
      - $P(x)$
    - $Q(f(c), g(c, g(c, d)))$
  - $\exists x P(x)$
    - $P(x)$



Logical degree of a formula

The *logical degree* of a formula is a nonnegative integer defined as follows:

$$\ell : Form \to \{0, 1, 2, \dots\}$$

If $A \in Con$ then

- the logical degree of an atomic formula A is 0
  $\ell(A) \rightleftharpoons 0;$

if $B \in Form$ and $C \in Form$ are arbitrary formulas,

- $\ell(\ \neg B\ ) \rightleftharpoons 1 + \ell(B);$
- $\ell(\ (B \circ C)\ ) \rightleftharpoons 1 + \ell(B) + \ell(C)$ where $\circ \in \{\wedge, \vee, \supset, \equiv\};$
- $\ell(\ QxB\ ) \rightleftharpoons 1 + \ell(B)$ where $Q \in \{\exists, \forall\}$ and $x \in Var$.

Functional degree of a term

The *functional degree* of a term is a nonnegative integer defined as follows:

$$\tilde{\ell} : Term \to \{0, 1, 2, \ldots\}$$

- If $t \in \mathcal{F}(0) \cup Var$ then
$$\tilde{\ell}(t) \;\rightleftharpoons\; 0$$
(the functional degree of an atomic term t is 0)

- if $f(t_1, t_2, \ldots, t_k) \in Term$ then
$$\tilde{\ell}(t) \;\rightleftharpoons\; 1 + \sum_{i=1}^{k} \tilde{\ell}(t_i)$$
(the functional degree of a compound term is 1 plus the sum of the functional degrees of the direct subterms)

Exercises

Terms

1. symbols used as name of functions:
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2. symbols used as name of predicates:
   - $\mathcal{P}(0)$ contains the propositional parameters,
   - $\mathcal{P}(k)$ where $k \geq 1$ contains the predicate parameters.

1. $Var \subseteq Term$;
2. $\mathcal{F}(0) \subseteq Term$;
3. if $f \in \mathcal{F}(k)$ for some $k \geq 1$ and $t_i \in Term$ for all $i = 1, \ldots, k$, then

$$f(t_1, t_2, \ldots, t_k) \in Term.$$

In this section (if we do not state something else) let the set the following: $\mathcal{F}(0) = \{c, d\}$, $\mathcal{F}(1) = \{f\}$, $\mathcal{F}(2) = \{g\}$, $\mathcal{P}(1) = \{P\}$, $\mathcal{P}(2) = \{Q\}$ and $Var = \{x, \ldots\}$

1. Which of the following are the terms of first-order language $L^{(1)}$? ☐ $f(g(d,d))$ ☐ $g(f(g(f(c), f(c))), x)$
   ☐ $g(d, g(c, f(x)))$ ☐ $g(d, f(c))$ ☐ $g(f(g(d,c)), d)$ ☐ $g(d, g(x, f(f(d))))$ ☐ $g(c, g(f(d), d))$
   ☐ $g(f(c), f(g(x, f(f(d)))))$ ☐ $g(d, f(f(c)))$ ☐ $f(g(f(x)))$ ☐ $f(f(d), f(f(c)))$ ☐ $g(f(g(c,x)), d)$
   ☐ $g(c, f(c))$ ☐ $g(f(d), f(f(f(x))))$ ☐ $g(d, g(g(c,x), c))$ ☐ $c(d)$ ☐ $g(g(g(d, f(c)), c), d)$
   ☐ $g(d, g(f(f(c)), d))$ ☐ $g(c, f(f(f(x))))$ ☐ $g(c, g(g(d, f(f(f(x)))), d))$ ☐ $P(g(d,d))$ ☐ $d = f(x)$
   ☐ $g(h(f(c), f(d)), d)$ ☐ $g(f(x), g(c, f(c)))$ ☐ $g(f(f(c)), d)$ ☐ $c = d$ ☐ $f(g(f(f(x)), d))$

Terms

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1. Which of the following are the terms of first-order language $L^{(1)}$? ■ $f(g(d,d))$ ■ $g(f(g(f(c), f(c))), x)$ ■ $g(d, g(c, f(x)))$ ■ $g(d, f(c))$ ■ $g(f(g(d,c)), d)$ ■ $g(d, g(x, f(f(d))))$ ■ $g(c, g(f(d), d))$ ■ $g(f(c), f(g(x, f(f(d)))))$ ■ $g(d, f(f(c)))$ □ $f(g(f(x)))$ □ $f(f(d), f(f(c)))$ ■ $g(f(g(c,x)), d)$ ■ $g(c, f(c))$ ■ $g(f(d), f(f(f(x))))$ ■ $g(d, g(g(c,x), c))$ □ $c(d)$ ■ $g(g(g(d, f(c)), c), d)$ ■ $g(d, g(f(f(c)), d))$ ■ $g(c, f(f(f(x))))$ ■ $g(c, g(g(d, f(f(f(x)))), d))$ □ $P(g(d,d))$ □ $d = f(x)$ □ $g(h(f(c), f(d)), d)$ ■ $g(f(x), g(c, f(c)))$ ■ $g(f(f(c)), d)$ □ $c = d$ ■ $f(g(f(f(x)), d))$

Terms

1. symbols used as name of functions:
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1. $Var \subseteq Term$;
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3. if $f \in \mathcal{F}(k)$ for some $k \geq 1$ and $t_i \in Term$ for all $i = 1, \ldots, k$, then

$$f(t_1, t_2, \ldots, t_k) \in Term.$$

The first-order language:

$$\mathcal{F}(0) = \{c, d, e\}, \quad \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\} \ \mathcal{F}(3) = \{h\}$$

$$\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\} \text{ and } Var = \{x, \ldots\}$$

2. Which of the following are the terms of first-order language $L^{(1)}$?

☐ $g(e, h(h(c, c, x), e, f(g(c, d))))$ ☐ x
☐ $h(f(h(x, f(f(f(d))), f(d))), f(c), d)$ ☐ $h(e, c, g(f(f(g(e, f(x)))), e))$ ☐ $g(g(f(d), f(d)), e)$
☐ $h(e, f(g(f(e), f(g(d, g(e, c))))), d)$ ☐ $g(d, c)$ ☐ $Q(h(d, d, f(c)), f(d))$ ☐ e ☐ $e = e$ ☐ $g(c, d)$ ☐ $g(h(c, c), e, f(f(d)))$ ☐ $f(g(g(e, g(c, c)), f(e)))$ ☐ $g(h(d, d, c), e)$
☐ $h(g(e, d), e, d)$ ☐ $g(f(d), e)$ ☐ $f(c)$ ☐ $h(g(g(e, g(c)), h(d, d, d)), h(e, f(d), e), d)$
☐ $e = f(d)$ ☐ $f(h(g(c, d), x, c))$ ☐ $h(h(c, d, d), f(e), d)$ ☐ $h(h(d, e, c), f(d), e)$

Terms

The first-order language:

$$\mathcal{F}(0) = \{c, d, e\}, \quad \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\} \ \mathcal{F}(3) = \{h\}$$

$$\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\} \text{ and } Var = \{x, \ldots\}$$

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$$f(t_1, t_2, \ldots, t_k) \in Term.$$

2. Which of the following are the terms of first-order language $L^{(1)}$?

■ $g(e, h(h(c,c,x), e, f(g(c,d))))$ ■ x
■ $h(f(h(x, f(f(f(d))), f(d))), f(c), d)$ ■ $h(e, c, g(f(f(g(e, f(x)))), e))$ ■ $g(g(f(d), f(d)), e)$
■ $h(e, f(g(f(e), f(g(d, g(e,c))))), d)$ ■ $g(d,c)$ □ $Q(h(d,d,f(c)), f(d))$ ■ e □ $e = e$
■ $g(c,d)$ □ $g(h(c,c), e, f(f(d)))$ ■ $f(g(g(e, g(c,c)), f(e)))$ ■ $g(h(d,d,c), e)$
■ $h(g(e,d), e, d)$ ■ $g(f(d), e)$ ■ $f(c)$ □ $h(g(g(e, g(c)), h(d,d,d)), h(e, f(d), e), d)$
□ $e = f(d)$ ■ $f(h(g(c,d), x, c))$ ■ $h(h(c,d,d), f(e), d)$ ■ $h(h(d,e,c), f(d), e)$

Terms

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$$f(t_1, t_2, \ldots, t_k) \in Term.$$

$L^{(1)}$ is a first order language where:

$$\mathcal{F}(0) = \{c, d\}, \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\},$$

$$\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\} \text{ and } Var = \{x, \ldots\}$$

Which of the formulae listed below are terms of the first order language?

3. ☐ $g(c, f(f(x)))$ ☐ $f(f(g(c, f(f(c)))))$ ☐ $g(f(c), g(d, d))$ ☐ $g(f(d), c)$ ☐ $g(g(f(f(x)), d), d)$ ☐ $f(g(c, f(f(c))))$ ☐ $g(f(d), g(c, c))$ ☐ $f(g(f(x), d))$ ☐ $f(f(g(f(c), d)))$ ☐ $g(d, f(g(c, f(d))))$ ☐ $g(g(g(d, f(x)), f(f(d))), c)$ ☐ $g(f(f(d)), d)$ ☐ $g(f(f(d)), f(x))$ ☐ c ☐ $g(f(c), c) = g(f(c), d)$ ☐ $g(g(f(d), x), d)$ ☐ $f(g(c, g(c, f(f(d)))))$ ☐ $g(f(g(f(f(d)), g(d))), g(f(d), d))$

Terms

1. symbols used as name of functions:
   - $\mathcal{F}(0)$ contains the name parameters,
   - $\mathcal{F}(k)$ where $k \geq 1$ contains the function parameters.
2. symbols used as name of predicates:
   - $\mathcal{P}(0)$ contains the propositional parameters,
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1. $Var \subseteq Term$;
2. $\mathcal{F}(0) \subseteq Term$;
3. if $f \in \mathcal{F}(k)$ for some $k \geq 1$ and $t_i \in Term$ for all $i = 1, \ldots, k$, then

$$f(t_1, t_2, \ldots, t_k) \in Term.$$

$L^{(1)}$ is a first order language where:

$$\mathcal{F}(0) = \{c, d\}, \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\},$$

$$\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\} \text{ and } Var = \{x, \ldots\}$$

Which of the formulae listed below are terms of the first order language?

3. ■ $g(c, f(f(x)))$ ■ $f(f(g(c, f(f(c)))))$ ■ $g(f(c), g(d, d))$ ■ $g(f(d), c)$ ■ $g(g(f(f(x)), d), d)$ ■ $f(g(c, f(f(c))))$ ■ $g(f(d), g(c, c))$ ■ $f(g(f(x), d))$ ■ $f(f(g(f(c), d)))$ ■ $g(d, f(g(c, f(d))))$ ■ $g(g(g(d, f(x)), f(f(d))), c)$ ■ $g(f(f(d)), d)$ ■ $g(f(f(d)), f(x))$ ■ c □ $g(f(c), c) = g(f(c), d)$ ■ $g(g(f(d), x), d)$ ■ $f(g(c, g(c, f(f(d)))))$ □ $g(f(g(f(f(d)), g(d))), g(f(d), d))$

Atomic formulae

$L^{(1)}$ is a first order language where:

$\mathcal{F}(0) = \{c, d\}, \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\},$

$\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\}$ and $Var = \{x, \ldots\}$

Which of the following are the atomic formulae of first-order language $L^{(1)}$?

4. ☐ $P(f(g(x,c)))$ ☐ $f(g(c,f(d)))$ ☐ $f(g(c,g(c,f(f(d)))))$ ☐ $P(f(d))$ ☐ $Q(g(g(d,f(x)),c),c)$ ☐ $c = f(g(d,f(d),c)))$ ☐ $g(d,x) = d$ ☐ $c = g(f(d),g(c,c))$ ☐ $g(g(f(g(g(g(d,c),d),c)),c),c) = c$ ☐ $P(g(g(g(x,g(f(x),c)),g(x,g(c,c))),d))$ ☐ $c = g(g(g(d,x),g(f(d),c)),c)$ ☐ $Q(g(c,g(f(g(g(d,d),g(c))),x)),f(d))$ ☐ $d = c$ ☐ $g(f(c),c) = c$ ☐ $Q(c,c)$ ☐ $c = c$ ☐ $Q(c,g(c,f(x)))$ ☐ $f(c) = f(c) \equiv \neg c = d$ ☐ $Q(g(c,c),x)$ ☐ $f(g(c,f(f(f(d)))))$ ☐ $P(g(c,g(x,c)))$ ☐ $Q(g(f(g(d,g(c,x))),c),c)$ ☐ $P(g(d,x))$ ☐ $f(g(c,f(f(c))))$ ☐ $\neg P(d) \wedge f(c) = f(g(c,d))$ ☐ $Q(g(g(c,c),c),c)$ ☐ $g(f(c),f(f(d))) = f(x)$ ☐ $\neg\neg P(c) \wedge P(c) \wedge Q(d,d)$ ☐ $Q(x,f(f(g(g(c),x))))$ ☐ $Q(f(f(x)),d)$ ☐ $f(g(c,c))$ ☐ $Q(c,f(d,x))$ ☐ $c = g(x,d)$

1. symbols used as name of functions:
   - $\mathcal{F}(0)$ contains the name parameters,
   - $\mathcal{F}(k)$ where $k \geq 1$ contains the function parameters.
2. symbols used as name of predicates:
   - $\mathcal{P}(0)$ contains the propositional parameters,
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Atomic formula:

1. $\mathcal{P}(0) \subseteq Form$;
2. $(t_1 = t_2) \in Form$ if
   - t_1 and t_2 are terms ($t_1 \in Term$ and $t_2 \in Term$)
3. $f(t_1, t_2, \ldots, t_k) \in Form$ if
   - $f \in \mathcal{P}(k)$ for some $k \geq 1$ and
   - $t_i \in Term$ for all $i = 1, \ldots, k$.

Atomic formulae

$L^{(1)}$ is a first order language where:

$$\mathcal{F}(0) = \{c, d\}, \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\},$$

$$\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\} \text{ and } Var = \{x, \ldots\}$$

Which of the following are the atomic formulae of first-order language $L^{(1)}$?

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Atomic formula:

1. $\mathcal{P}(0) \subseteq Form$;
2. $(t_1 = t_2) \in Form$ if
   - t_1 and t_2 are terms ($t_1 \in Term$ and $t_2 \in Term$)
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   - $f \in \mathcal{P}(k)$ for some $k \geq 1$ and
   - $t_i \in Term$ for all $i = 1, \ldots, k$.

Formulae

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Formulas:

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4. $\neg A \in Form$ if A is a formula ($A \in Form$);
5. $(A \circ B) \in Form$ if
   - A and B are formulas ($A \in Form$, $B \in Form$)
   - $\circ$ is a binary connective ($\circ \in \{\wedge, \vee, \supset, \equiv\}$);
6. $QxA \in Form$ if
   - Q is a quantifier ($Q \in \{\exists, \forall\}$),
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$L^{(1)}$ is a first order language where:

$$\mathcal{F}(0) = \{c, d\}, \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\},$$

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Which of the following are the formulae of first-order language $L^{(1)}$?

5. ☐ $\neg P(d) \wedge f(c) = f(g(c,d))$ ☐ $\forall x \neg c = f(d)$ ☐ $d = c \wedge P(d) \equiv \neg P(c)$ ☐ $\forall x Q(c,d)$
   ☐ $Q(c,c) \vee P(c) \wedge c = c \wedge P(c) \wedge c = d \equiv Q(d,c)$ ☐ $f(e) = f(c)$ ☐ $\neg P(c) \equiv P(c) \wedge c = c \equiv \exists x Q(x,x)$
   ☐ $c = g(c,c) \equiv c = d \vee Q(c,d) \wedge Q(c,d) \supset c = d$ ☐ $Q(g(c), f(g(d, g(c,c))))$ ☐ $g(f(c), c) = g(f(c), d)$

Formulae

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Which of the following are the formulae of first-order language $L^{(1)}$?

5. ■ $\neg P(d) \wedge f(c) = f(g(c,d))$ ■ $\forall x \neg c = f(d)$ ■ $d = c \wedge P(d) \equiv \neg P(c)$ ■ $\forall x Q(c,d)$

■ $Q(c,c) \vee P(c) \wedge c = c \wedge P(c) \wedge c = d \equiv Q(d,c)$ □ $f(e) = f(c)$ ■ $\neg P(c) \equiv P(c) \wedge c = c \equiv \exists x Q(x,x)$

■ $c = g(c,c) \equiv c = d \vee Q(c,d) \wedge Q(c,d) \supset c = d$ □ $Q(g(c), f(g(d, g(c,c))))$ ■ $g(f(c), c) = g(f(c), d)$

Formulae

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$L^{(1)}$ is a first order language where:

$\mathcal{F}(0) = \{c, d\}, \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\},$

$\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\}$ and $Var = \{x, \ldots\}$

6. Which of the following are the formulae of first-order language $L^{(1)}$?

☐ $Q(d,c)$ ☐ $Q(c,d) \wedge P(x,c) \supset Q(c,c) \vee P(c) \wedge c = c \vee P(c)$ ☐ $Q(d,d) \supset d = \exists x g(x,d)$
☐ $\neg Q(d,d)$ ☐ $c = c \equiv f(c) = g(c, f(c)) \vee P(c)$ ☐ $\neg P(d)$ ☐ $P(g(c,c))$ ☐ $P(f(c)) \vee Q(c,d) \equiv \neg P(g(g(c,d), g(c,d)))$ ☐ $P(c) \wedge P(c) \supset \neg P(c) \vee Q(d,d)$ ☐ $\neg g(c, g(c,d)) = c$ ☐ $d = d \equiv \exists d P(d)$
☐ $P(c) \& Q(d,x)$ ☐ $d = c \supset d = g(c,c)$ ☐ $c = f(g(f(c), g(c,d)))$ ☐ $c = \neg c \supset P(c) \supset Q(c,c) \wedge Q(d,d)$ ☐ $f(c) = f(c) \equiv \neg c = d$ ☐ $\neg d = c$ ☐ $P(f(f(d))) \vee d = d$ ☐ $\neg Q(h(d,d,f(c)), f(d))$
☐ $\neg P(f(d))$ ☐ $P(g(f(g(c,d)), c))$ ☐ $P(d)$ ☐ $\neg c = f(c) \supset g(c,c) = d$ ☐ $\exists c P(c) \equiv Q(f(c), c)$
☐ $f(f(d)) = g(c, f(f(c)))$ ☐ $\neg g(c,c) = d$ ☐ $c = c$ ☐ $g(g(c, g(c,d)), g(g(c,c), f(c))) = c$
☐ $\neg\neg P(c) \supset Q(c, g(c,d)) \equiv d = c$ ☐ $d = c$ ☐ $Q(c, f(d)) \wedge Q(f(c), f(g(c, g(c,c))))$ ☐ $Q(c, f(c)) \vee$

Formulae

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6. Which of the following are the formulae of first-order language $L^{(1)}$?

$\blacksquare$ $Q(d,c)$ $\square$ $Q(c,d) \wedge P(x,c) \supset Q(c,c) \vee P(c) \wedge c = c \vee P(c)$ $\square$ $Q(d,d) \supset d = \exists x g(x,d)$
$\blacksquare$ $\neg Q(d,d)$ $\blacksquare$ $c = c \equiv f(c) = g(c,f(c)) \vee P(c)$ $\blacksquare$ $\neg P(d)$ $\blacksquare$ $P(g(c,c))$ $\blacksquare$ $P(f(c)) \vee Q(c,d) \equiv \neg P(g(g(c,d),g(c,d)))$ $\blacksquare$ $P(c) \wedge P(c) \supset \neg P(c) \vee Q(d,d)$ $\blacksquare$ $\neg g(c,g(c,d)) = c$ $\square$ $d = d \equiv \exists d P(d)$
$\square$ $P(c) \& Q(d,x)$ $\blacksquare$ $d = c \supset d = g(c,c)$ $\blacksquare$ $c = f(g(f(c),g(c,d)))$ $\square$ $c = \neg c \supset P(c) \supset Q(c,c) \wedge Q(d,d)$ $\blacksquare$ $f(c) = f(c) \equiv \neg c = d$ $\blacksquare$ $\neg d = c$ $\blacksquare$ $P(f(f(d))) \vee d = d$ $\square$ $\neg Q(h(d,d,f(c)),f(d))$
$\blacksquare$ $\neg P(f(d))$ $\blacksquare$ $P(g(f(g(c,d)),c))$ $\blacksquare$ $P(d)$ $\blacksquare$ $\neg c = f(c) \supset g(c,c) = d$ $\square$ $\exists c P(c) \equiv Q(f(c),c)$
$\blacksquare$ $f(f(d)) = g(c,f(f(c)))$ $\blacksquare$ $\neg g(c,c) = d$ $\blacksquare$ $c = c$ $\blacksquare$ $g(g(c,g(c,d)),g(g(c,c),f(c))) = c$
$\blacksquare$ $\neg\neg P(c) \supset Q(c,g(c,d)) \equiv d = c$ $\blacksquare$ $d = c$ $\blacksquare$ $Q(c,f(d)) \wedge Q(f(c),f(g(c,g(c,c))))$ $\blacksquare$ $Q(c,f(c)) \vee$

Construction tree

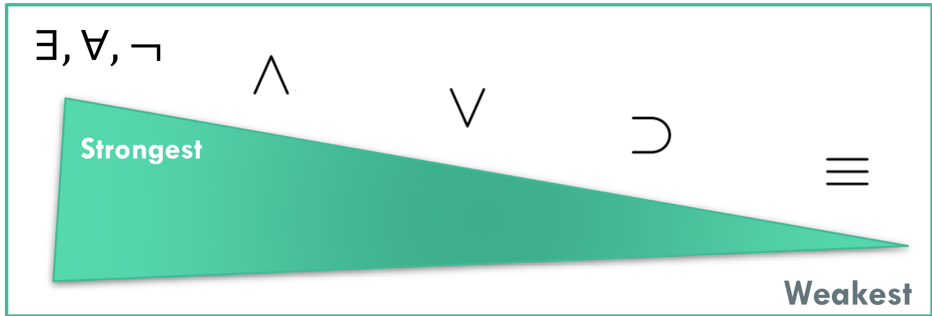
25. Give the construction tree of formula $\forall x P(x) \wedge Q(f(c), g(c, g(c, d))) \vee \exists x P(x)$!

$$\forall x P(x) \wedge Q(f(c), g(c, g(c, d))) \vee \exists x P(x)$$

$$\forall x P(x) \wedge Q(f(c), g(c, g(c, d))) \qquad \exists x P(x)$$

$$\forall x P(x) \qquad Q(f(c), g(c, g(c, d))) \qquad P(x)$$

$$P(x)$$



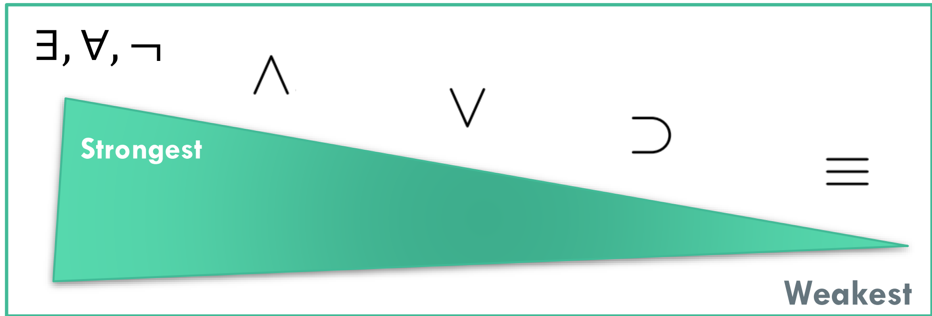
Direct sub-formulae

5. Give the direct sub-formulae of formula $\forall x P(x) \land Q(f(c), g(c, g(c, d))) \lor \exists x P(x)$!

6. Give the direct sub-formulae of formula $\forall x (P(x) \land Q(f(c), g(c, g(c, d))) \lor \exists x P(x))$!

7. Give the direct sub-formulae of formula $Q(c, g(c, g(c, d))) \lor \exists x (Q(g(c, c), d) \land Q(d, g(c, c)))$!

8. Give the direct sub-formulae of formula $P(g(d, d))$!

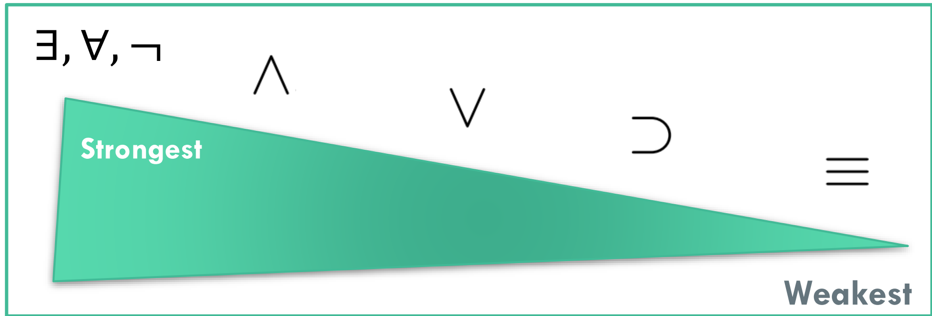


Sub-formulae

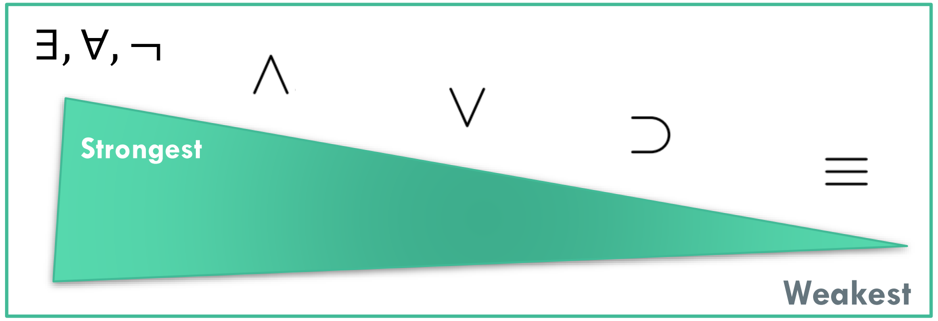
1. Give all the sub-formulae of formula $Q(c, g(c, g(c, d))) \vee \exists x Q(g(c, c), d) \wedge Q(d, g(c, c))$!

2. Give all the sub-formulae of formula $\forall x P(x) \wedge Q(f(c), g(c, g(c, d))) \vee \exists x P(x)$!

21. Give all the sub-formulae of formula $Q(c, g(c, g(c, d))) \vee \exists x Q(g(c, c), d) \wedge Q(d, g(c, c))$!



Construction trees



1. Give the construction tree of formula $\forall x P(x) \wedge Q(f(c), g(c, g(c, d))) \vee \exists x P(x)$!

2. Give the construction tree of formula

$$P(f(c)) \vee Q(c, d) \equiv \forall x \neg P(g(g(x, d), g(c, x)))!$$

3. Give the construction tree of formula

$$P(c) \supset P(c) \vee c = d \equiv \neg P(c) \wedge P(d) \wedge P(c) \wedge c = g(c, f(c)) \supset d = c!$$